DO NOW — RECALL

Procedures, functions and parameters

Name: _____ Class: _____ Date: _____

1. A local x=4 shadows a field x=9. Which is printed inside the method?

2. What is a base case?

3. What is a[2] for int[] a = {3,5,8,13}?

4. Starting x=1, apply x=x*2 three times. List the results.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

4. The local variable shadows the field.

2. Model answer

The condition that stops further recursive calls.

3. Model answer

8, because arrays start at index zero.

4. Model answer

2, 4, 8.

Vocabulary — know these terms

procedure — A subroutine that performs actions and does not return a result.

function — A subroutine that returns a value.

parameter — A named input in the subroutine declaration.

argument — The actual value provided by the caller.

return — Finish a function and pass its result to the caller.

NEW LEARNING

Read, annotate and explain

CALL

The caller supplies argument 6 to Double.

PARAMETER

Parameter value receives 6, then value * 2 evaluates to 12.

RETURN

The function returns 12. Console.WriteLine displays the result.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Trace AddVAT(50), which returns price + price / 5.

Predict first

The argument 50 becomes local parameter price = 50.

Model the result

Integer division 50 / 5 = 10; the returned value is 60.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

A function Square(n) returns n*n. Find Square(5).

Question 2

Write a C# function IsEven(int n) that returns true when n is even.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain why static int Show(int x) { Console.WriteLine(x); } is invalid. Correct it.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A C# method with return type void must return an integer.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.